

JEE MAIN REVISION MASTER: CIRCULAR MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 73 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Circular Motion — Standard Calculations & Problem Archetypes	58 PYQs	Quantitative Analysis
Very High	Circular Motion — Core Principles & Governing Laws	20 PYQs	Core Principle
High	Circular Motion — Applied Cases, Reactions & Traps	9 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Circular Motion — Standard Calculations & Problem Archetypes	$N = m\omega^2 r$, $T + mg \sin(30^\circ) = \frac{mv_P^2}{r}$, $\frac{1}{2}mv_1^2 = \frac{1}{2}mv_2^2 + mg2l$
Circular Motion — Core Principles & Governing Laws	$T - mg \cos \theta = \frac{mv^2}{R}$, $T + mg \sin(30^\circ) = \frac{mv_P^2}{r}$, $F_{\text{friction}} = \mu mg$.
Circular Motion — Applied Cases, Reactions & Traps	$f = Mg$, $R = \frac{mv^2}{R \cos \theta}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Circular Motion — Standard Calculations & Problem Archetypes (Very High Yield — 58 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Circular Motion.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A car moving with a speed of 54km/h takes a turn of radius 20 m . A simple pendulum is suspended from the ceiling of the car. Determine th..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "In case of vertical circular motion of a particle by a thread of length r if the tension in the thread is zero at an angle 30° sh..." [Ans: A]
- [2026 Morning Shift] Q#2 (MCQ): "A smooth inclined plane ends in a vertical circular loop, as shown in the figure. A small body is released from height ■ as shown. If the ..." [Ans: B]

3.2 Circular Motion — Core Principles & Governing Laws (Very High Yield — 20 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Circular Motion.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Circular Motion.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Circular Motion PYQs.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A car moving with a speed of 54km/h takes a turn of radius 20 m . A simple pendulum is suspended from the ceiling of the car. Determine th..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "In case of vertical circular motion of a particle by a thread of length r if the tension in the thread is zero at an angle 30° sh..." [Ans: A]
- [2026 Morning Shift] Q#2 (MCQ): "A smooth inclined plane ends in a vertical circular loop, as shown in the figure. A small body is released from height ■ as shown. If the ..." [Ans: B]

3.3 Circular Motion — Applied Cases, Reactions & Traps (High Yield — 9 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Circular Motion.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "A large drum having radius R is spinning around its axis with angular velocity ω , as shown in figure. The minimum value of ω ..." [Ans: D]
- [2023 Evening Shift] Q#2 (MCQ): "A stone of mass 1kg is tied to end of a massless string of length 1m . If the breaking tension of the string is 400N , then maximum li..." [Ans: N/A]
- [2023 Morning Shift] Q#2 (MCQ): "A car is moving with a constant speed of 20 m/s in a circular horizontal track of radius 40 m. A bob is suspended from the roof of the car b..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A car moving with a speed of 54km/h takes a turn of radius 20 m . A simple pendulum is suspended from the ceiling of the car. Determine the angle made by the string of the pendulum with the vertical during the turning. (Take $g=10\text{m/s}^2$)

- (A) $\tan^{-1}(0.5)$
- (B) $\tan^{-1}(0.75)$
- (C) $\tan^{-1}(1.125)$ [Correct]
- (D) $\tan^{-1}(0.25)$

Correct Answer: C

Solution Analysis: The speed of the car is given in km/h . We must convert it to m/s to match the units of gravity and radius. $v=54 \times (5 / 18)=3 \times 5=15\text{m/s}$ When the car takes a turn, from the perspective of an observer inside the car, the pendulum bob experiences gravity (mg) which is acting vertically downwards and centrifugal force (F_c) a pseudo-force acting horizontally outwards due to the circular motion. $F_c=\frac{mv^2}{r}$ Where m is mass, v is velocity, and r is radius. Let the string make an angle θ with the vertical. The tension T in the string can be resolved into

Benchmark Problem 2: JEE Main 2026 (Online) 21st January Evening Shift (Q#2)

A large drum having radius R is spinning around its axis with angular velocity ω , as shown in figure. The minimum value of ω so that a body of mass M remains stuck to the inner wall of the drum, taking the coefficient of friction between the drum surface and mass M as μ , is :

- (A) $\sqrt{(2g / \mu R)}$
- (B) $\sqrt{(g / 2\mu R)}$
- (C) $\sqrt{(\mu g / R)}$
- (D) $\sqrt{(g / \mu R)}$ [Correct]

Correct Answer: D

Solution Analysis: When the drum rotates, the mass M experiences forces : 1. Weight (Mg) acts vertically downwards. 2. Normal reaction (N) acts horizontally toward the centre of the drum. This provides the necessary centripetal force. 3. Frictional force (f) acts vertically upwards, opposing the tendency of the mass to slide down due to gravity. For the body to remain rest with respect to the cylindrical surface, the vertical forces must be balanced : $f=Mg$ The horizontal normal force provides the centripetal acceleration ($a_c=R\omega^2$) : $N=MR\omega^2$ The frictional force (f) is a self-adjusting